List of materials used in eVTOLS:

The selection of composite materials for most of the eVTOL projects is evident and not surprising given the characteristics that enable the use of this. Among the most important of these is the outstanding strength-to-weight ratio that makes it possible to improve the performance of the aircraft by improving its range, autonomy, and power requirements, which translates into the ability to carry passengers or cargo in a more economical and cost-effective manner. However, the use of composite materials is not the most economical option compared to other traditional methods since it must be done with all the safety measures and have a constant and complex maintenance to the structure. However, with this option it is possible to simplify the supply chain by using a narrow range of materials, which are safe and can be certified, in addition, the scalability or production in large quantities of this type of structures is feasible. These types of materials also adapt in various ways to the production of the required part, whether by compression molding, pultrusion, injection, infusion, Sheet Metal Compound, etc.

An example of companies favoring this option is Archer, BETA and Joby. In general, this material option is often used in the manufacture of fuselages, tail booms and wings. Alternatives to this material are fiberglass and Kevlar reinforcements. In the eVTOLS materials market, composites can make up about 70% of the structure. Some of the references in the field of composite materials for UAM are Aria Group, Toray Advanced Composites, Hexcel and Autoscale CNC.

In addition to these, they can already vary for cabin interiors depending on what is desired as well as specific components such as engines, batteries, and other parts other than the aircraft structure.

Write considerations for manufacturing that are different from traditional:

The use of composite materials brings with it the acquisition of good techniques and practices, e.g.:

1. Material selection and handling: Since they are composite materials, the combination between the matrix and the reinforcement must be done carefully to obtain the necessary strength and stiffness. In addition to the correct use of resin and fiber orientation.
2. Manufacturing techniques: Given the variety of processes, knowing the options and advantages and disadvantages of each one is of vital importance for the manufacture of parts or pieces.
3. Quality control should also be a strength in these processes, since bad practices or defective parts can lead to component failure and unexpected behavior against the estimated loads.
4. Safety and health: These materials can generate harmful health effects during their manufacture. For this reason, the use of equipment is necessary as well as the correct supervision of regulatory compliance.

Taking these factors into account, it is possible to achieve the advantages of composite materials without the possible failures at the time of manufacturing, which reduces time and costs for the production chain.